

Unit - I

30/9/25 Running notes:

AC: Continuous with time, Continuous with Amp

DC: Discrete Discrete

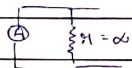
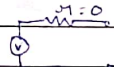
Discrete circuits Integrated circuits

We design, hand pick & Components are micro & milli connect

Advantage of DC: easy to store, transmit, less noise
b/w 2 points

Voltage source: Provides a const volt, irrespective of the current flowing ^{drawn} whose internal $r_i = 0$

Current source: const I irrespective of voltage across the circuit whose internal $r_i = \infty$



Short circuit: 2 points at same pot, $V=0$, $I=0$
 $R=0$, $L=0$, $C=\infty$

Open circuit: $R=\infty$, $L=\infty$, $C=0$, $I=0$

3 terminal devices

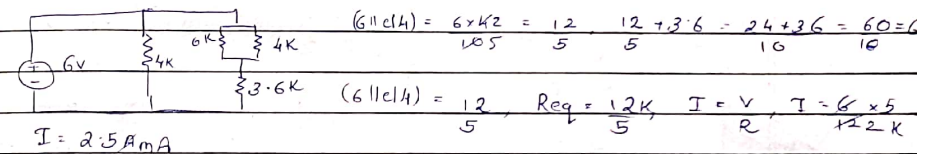
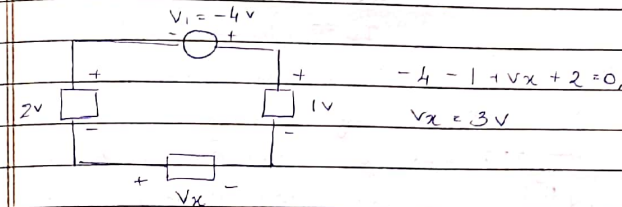
BJT

npn < > pnp

FET

↳ MOSFET → normally off
↳ Enhancement - n
↳ Depletion p

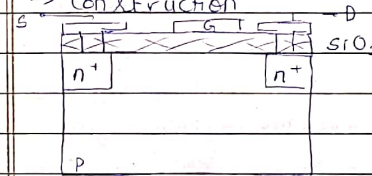
Problems:



3 terminal devices

n Type Metal oxide field effect Transistor

→ Construction



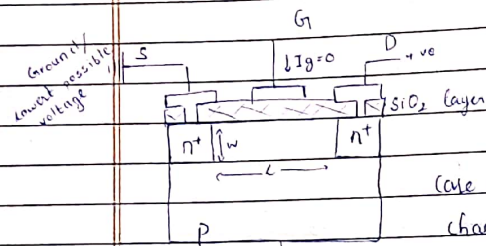
Application of BJT & MOSFET

6/10/25

N-channel

(case i) $V_{GS} = 0, V_D = 0, V_S = 0$

$I_G = 0$ Reverse biased diode / no
biase diode b/w sto Body & D to
body, $I_D = 0$

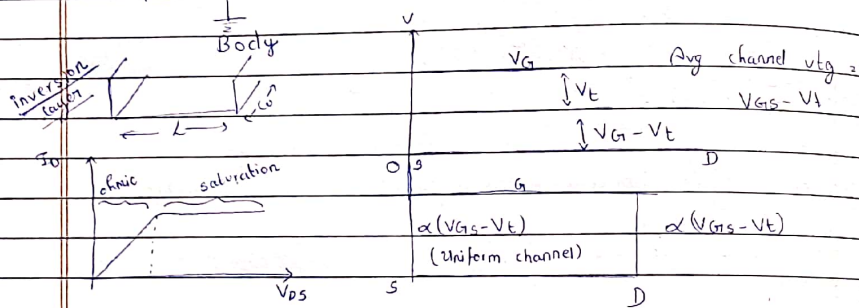


(case ii) $V_{GS} = +ve, V_D, V_S = 0$

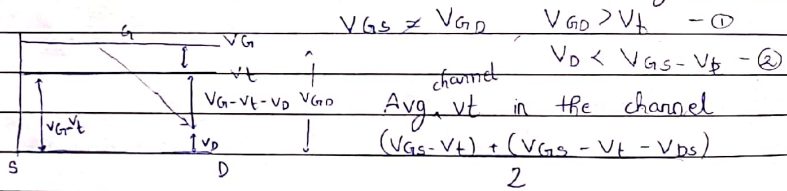
accumulation of holes, $I_D = 0$

(case iii) $V_{GS} = +ve, V_D = V_S = 0$

channel formed when $V_{GS} > V_t$



(case iv) $V_{GS} > V_t, V_S = 0V, V_D = +ve (mv \Rightarrow$ ohmic region)

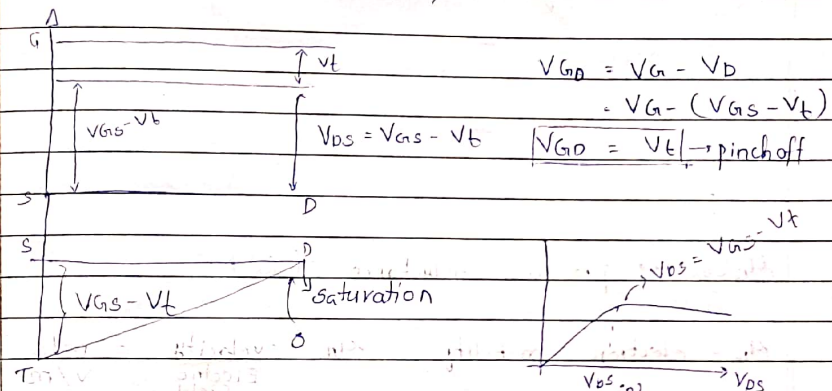


$$I_D = \frac{mA}{4A} = \frac{2V_{GS} - 2V_t - V_{DS}}{2} = V_{GS} - V_t - \frac{V_{DS}}{2}$$

$$V_{DS} < V_{GS} - V_t$$

$$V_D < V_G - V_t$$

(Case v) $V_{GS} > V_t, V_S = 0, V_{DS} = V_D = V_{GS} - V_t$

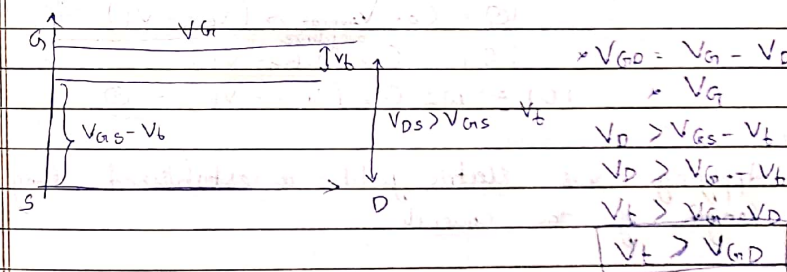


Avg voltage in the channel : $\frac{(V_{GS} - V_t) + 0}{2}$

$$= \frac{V_{GS} - V_t}{2}$$

(Case vi) $V_{GS} > V_t, V_S = 0, V_D \geq V_{GS} - V_t$

Current stops after pinch off, but $I_D = 0 \Rightarrow$
uniform : channel (case iii) so it becomes open, but
 V_D is applied, again pinch off. to & for motion
results in current not $\uparrow$ nor $\downarrow$



Conditions for saturation

in case v)

$$V_{GS} > V_t$$

$$V_{DS} > V_{GS} - V_t$$

$$V_{GD} > V_t$$

case vi)

$$V_{GS} > V_t$$

$$V_{DS} > V_{GS} - V_t$$

$$V_{GD} < V_t$$

Conditions for Saturation Current

$\mu_n C_{ox} \rightarrow$ process conductance (K_n')

μ_n - electron mobility $\mu_n = \frac{\text{velocity}}{\text{Electric field}} = \frac{m/s}{V/m}$

$$\mu_n = \frac{m^2}{V \cdot s}$$

$$C_{ox} = \frac{F}{m^2}$$

$$\Rightarrow \mu_n C_{ox} = \frac{m^2}{V \cdot s} \times \frac{F}{m^2} = \frac{F}{V \cdot s} = \frac{A}{V \cdot V} = \frac{A}{V^2}$$

charge in channel is given by

$$|Q| \propto V$$

$$|Q| = C \cdot V$$

$$|Q| = C_{ox} V_{\text{overdrive}} \rightarrow (V_{GS} - V_t)$$

$$|Q| = C_{ox} [V_{GS} - V_t]$$

$$|Q| = W L C_{ox} [V_{GS} - V_t] \quad - (1)$$

Applying V_{DS} Electric field is established along the length of the channel

$$|E| = \frac{V_{DS}}{L} \quad - (2)$$

$$\text{velocity} = \frac{\text{dist}}{\text{time}} = \frac{L}{t} \quad - (3)$$

MOSFET at saturation behaves as a current source

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$$I_D = \frac{|Q|}{t} = \frac{W L C_{ox} [V_{GS} - V_t]}{t} = \frac{W L C_{ox} [V_{GS} - V_t] \times \text{velocity}}{L}$$

from (3): $t = \frac{L}{\text{velocity}}$ velocity = $\mu_n |E|$
velocity = $\mu_n \cdot \frac{V_{DS}}{L}$

$$I_D = \mu_n C_{ox} \frac{W}{L} [V_{GS} - V_t] \cdot V_{DS} \Rightarrow (A)$$

Avg channel E

i) for ohmic condition, Avg $v = \frac{V_{GS} - V_t - V_{DS}}{2}$

$$I_D = \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_t - \frac{V_{DS}}{2}) \cdot V_{DS}$$

$$I_D = K_n' \left[(V_{GS} - V_t) V_{DS} - \frac{V_{DS}^2}{2} \right]$$

for small V_{DS} values (mV) $\frac{V_{DS}^2}{2} \ll 1$, $\therefore$ neglect the 2nd term

$$I_D = \mu_n C_{ox} \frac{W}{L} [V_{GS} - V_t] V_{DS}$$

$$I_D = I = \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_t) V_{DS}$$

$$R = \frac{1}{\mu_n C_{ox} \frac{W}{L} (V_{GS} - V_t)}$$

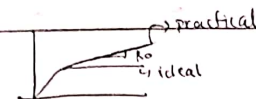
ii) for $V_{DS} = V_{GS} - V_t$; Avg channel vltg = $\frac{V_{GS} - V_t}{2}$ - (C)
put (C) in (A)

$$I_D = K_n' (V_{GS} - V_t) V_{DS}$$

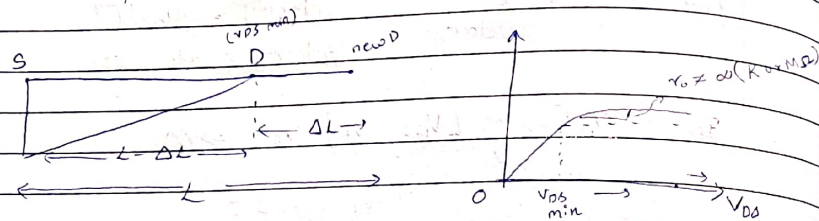
$$I_D = K_n' V_{DS} (V_{GS} - V_t); \text{ for } V_{DS} = V_{GS} - V_t$$

$$I_D = K_n' (V_{GS} - V_t)^2$$

ideally $R = \infty$



Channel length modulation (non-ideal behaviour / finite o/p resistance in saturation)



$$I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_t)^2$$

$$I_{D(sat)} = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_t)^2$$

Due to channel length modulation length of the channel changes from \$L\$ to \$L - \Delta L\$

$$I_{DCLM} = \frac{1}{2} \mu_n C_{ox} \frac{W}{L - \Delta L} (V_{GS} - V_t)^2$$

$$I_{DCLM} = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} \times \frac{L}{L - \Delta L} (V_{GS} - V_t)^2$$

$$I_{DCLM} = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} \left(\frac{1}{1 - \Delta L/L} \right) (V_{GS} - V_t)^2 \quad (\text{Let } x = \Delta L/L)$$

$$I_{DCLM} = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (1 - x)^{-1} (V_{GS} - V_t)^2$$

$$I_{DCLM} = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (1 + \lambda) (V_{GS} - V_t)^2 \quad [(1 - x)^{-1} = 1 + \lambda]$$

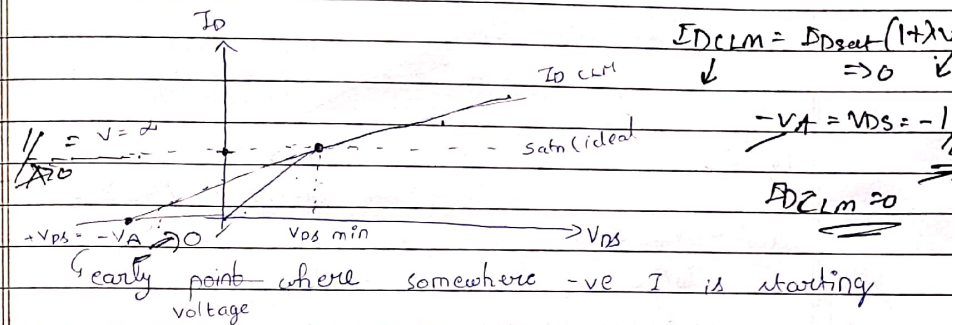
$$\text{Let } \Delta L \propto V_{DS} \Rightarrow \Delta L = \lambda' V_{DS}$$

$$I_{DCLM} = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} V_{GS}^2 \left[1 + \frac{\lambda' V_{DS}}{L} \right]$$

$$I_{DCLM} = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} V_{GS}^2 \left[1 + \lambda V_{DS} \right] \quad \lambda = \frac{\lambda'}{L}$$

$$I_{D(CL)} = I_{D(sat)} (1 + \lambda V_{DS})$$

$$V_A = \frac{1}{\lambda}$$



$$I_{DCLM} = 0 \quad \text{when} \quad V_{DS} = -1/\lambda$$

If there is CLM \$\lambda\$ = some value, if there is no CLM, \$\lambda = 0\$, \$\Rightarrow V_A = 1/0 = \infty\$

if \$\lambda = 0\$, \$V_A = \infty\$, \$r_o = \infty\$, \$I_D = I_{D(sat)}\$

if \$\lambda \neq 0\$, \$-V_A = +1/\lambda\$, \$I_D = I_{DCLM}\$, \$r_o \neq \infty\$ finite op resistance

diff \$I_{DCLM}\$ with \$V_{DS}\$

$$\partial I_{DCLM} = I_{D(sat)} (\lambda)$$

$$\partial V_{DS}$$

$$r_o = \frac{\partial V_{DS}}{\partial I_{DCLM}} = \frac{1}{\lambda I_{D(sat)}}$$

$$r_o = \frac{V_A}{I_{D(sat)}}$$

In ideal MOSFET, incremental o/p resistance \$r_o = \infty\$ for incremental \$V_{DS}\$ & \$I_{D(sat)} = I_{D(sat)}\$

In practical mosfet incremental o/p resistance
 $r_o = \frac{1}{\lambda I_{Dsat}} = \frac{V_A}{I_{Dsat}} \text{ (K}\Omega \text{ or M}\Omega\text{)}$ for incremental

$V_{DD} \dots I_D = I_{Dsat}$

NMOS

PMOS

1) cutoff: $V_{GS} \leq V_t$
 $I_D = 0$

1) Cutoff: $V_{GS} \geq V_t$
 $I_D = 0$

2) Linear: ohmic / Triode
 $V_{GS} > V_t$

$V_{DS} < V_{GS} - V_t$

$V_{GD} > V_t$

$I_D = I_{Dohmic}$

$= K'_n \left[(V_{GS} - V_t) V_{DS} - \frac{V_{DS}^2}{2} \right]$

3) Saturation

$V_{GS} > V_t$

$V_{DS} \geq V_{GS} - V_t$

$V_{GD} \leq V_t$

$I_D = I_{Dsat} \text{ (ideal)}$

$I_D = I_{Dsat} \text{ (practical)}$

2) Linear

$V_{GS} < |V_t|$

$V_{DS} > V_{GS} - V_t$

$V_{GD} < |V_t|$

$I_D = I_{Dohmic}$

3) Saturation

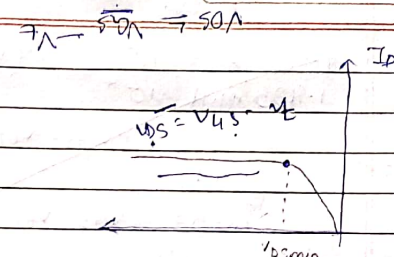
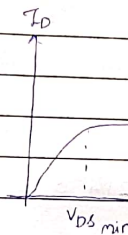
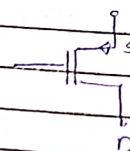
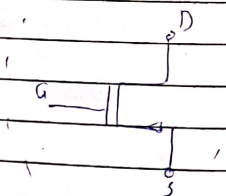
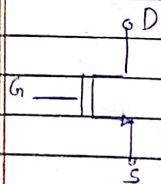
$V_{GS} < V_t$

$V_{DS} \leq V_{GS} - V_t$

$V_{GD} \geq V_t$

$I_D = I_{Dsat} \text{ (ideal)}$

$I_D = I_{Dsat} \text{ (practical)}$



Q1. Calculate V_{GS} & V_{DSmin} required to operate MOSFET transistor in saturation. $I_D = 100 \mu A$, $\frac{W}{L} = 10$
 $\mu_n C_{ox} = 195 \mu A/V^2$, $V_t = 1V$

$I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_t)^2$

$100 \times 10^{-6} = \frac{1}{2} \times 195 \times 10^{-6} \times 10 (V_{GS} - 1)^2$

$20 = (V_{GS} - 1)^2$, $(V_{GS} - 1)^2 = 0.1025$, $V_{GS} - 1 = 0.3201$

$V_{GS} = 1.3201 V$

V_{DS} at pinchoff $V_{DS} = V_{GS} - V_t$
 $= 0.3201$

Q2. An NMOS transistor is fabricated with $L = 180nm$
 $W = 2.4\mu m$, given: $C_{ox} = 8.6 fF/(\mu m)^2$, $\mu_n = 450 cm^2/V \cdot s$
 $V_t = 0.5$. Find V_{GS} & V_{DS} such that the MOSFET operates at edge of sat, with $I_D = 100 \mu A$

Given $\frac{W}{L} = \frac{2.4 \times 10^{-6}}{180 \times 10^{-9}} = \frac{1000}{90}$, $\mu_n = 450 \times 10^{-4} m^2/V \cdot s$

$C_{ox} = \frac{8.6 \times 10^{-15} F}{10^{-12} m^2} = 8.6 \times 10^{-3} F/m^2$, $I_D = 100 \times 10^{-6} A$

$100 \times 10^{-6} = \frac{1}{2} \times 450 \times 10^{-4} \times 8.6 \times 10^{-3} \times \frac{1000}{90} (V_{GS} - V_t)^2$

$$\frac{100 \times 10^{-6}}{1.935 \times 11.1 \times 10^{-7}} = (V_{GS} - V_t)^2$$

$$\frac{1000}{21.49} = (V_{GS} - V_t)^2$$

$$46.53 = (V_{GS} - V_t)^2, V_{GS} = 6.821 + 0.5$$

$$V_{GS} = 7.321$$

$$V_{DS} = V_{GS} - V_t, V_{DS} = 6.821$$

$$0.00465 \times 10 = (V_{GS} - V_t)^2$$

$$0.0465 = (V_{GS} - V_t)^2$$

$$0.2156 = V_{GS} - V_t$$

$$V_{GS} = 6.7156$$

$$V_{DS} = 0.2156$$

ii) If V_{GS} is kept const, find V_{DS} that results in $I_D = 50 \mu A$

$$V_{GS} = 6.7156$$

$$I_D = \frac{\mu_n C_{ox} W}{L} (V_{DS})^2$$

$$50 \times 10^{-6} = \frac{1 \times 450 \times 10}{2}$$

$$I_D = K'_n [(V_{GS} - V_t) V_{DS} - \frac{V_{DS}^2}{2}]$$

$$50 \times 10^{-6} = 450 \times 10^{-4} \cdot 8.6 \times 10^{-3} \times 11.1 (6.7156 - 0.5) V_{DS} - \frac{V_{DS}^2}{2}$$

$$50 \times 10^{-6} = 42957 \times 10^{-7} ($$

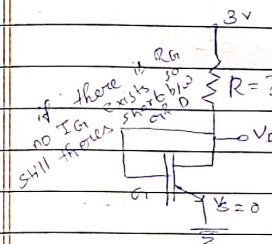
$$\frac{500}{42957} = (0.2156 V_{DS} - \frac{V_{DS}^2}{2})$$

$$0.0116 = (0.2156 V_{DS} - \frac{V_{DS}^2}{2})$$

$$0.0232 = 0.4312 V_{DS} - \frac{V_{DS}^2}{2}$$

$$V_{DS}^2 - 0.4312 V_{DS} + 0.0232 = 0$$

② Design the circuit to obtain $I_D = 80 \mu A$, find required value of R , V_{DS} voltage at drain if $V_t = 0.6 V$, $\mu_n C_{ox} W = 200 \mu A/V^2 \times 5$.



$$A_1 V_{GS} = V_D$$

$$V_{DS} = V_{GS} - V_t$$

$$V_{DS} > V_{GS} - V_t \Rightarrow \text{saturation}$$

$$I_D = \frac{K'_n}{2} (V_{GS} - V_t)^2$$

$$\frac{80 \times 10^{-6} \times 2}{200 \times 10^{-6} \times 5} = (V_{GS} - V_t)^2, \frac{0.8}{5} = \frac{(V_{GS} - V_t)^2}{V_{GS}}$$

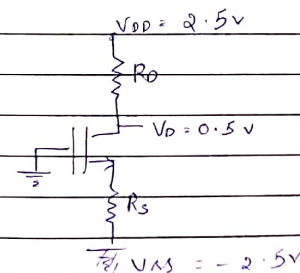
$$0.8444 = V_{DS} - V_t, V_{DS} = 1.4944 V$$

$$0.4 V_{DS} = 0.661$$

$$V_{GS} = 1 V$$

$$R = \frac{3 - V_D}{I_D}, R = \frac{3 - 1}{80 \times 10^{-6}} = 0.025 \times 10^6 = 25 K\Omega$$

③ Design circuit so that transistor operates at $I_D = 0.4 mA$
 $V_D = 0.5 V$, $V_t = 0.7 V$, $\mu_n C_{ox} = 100 \mu A/V^2$, $W = 32$
 $L = 0$



Assume transistor is in saturation

$$I_D = \frac{K'_n}{2} (V_{GS} - V_t)^2$$

$$I_D = 0.4 \times 10^{-3} = \frac{100 \times 10^{-6} \times 32}{2} (V_{GS} - V_t)^2$$

$$2.5 \times 10^{-4} \times 10^{-3} = 10^{-6} (V_{GS} - V_t)^2$$

$$2.5 \times 10^{-1} = (V_{GS} - V_t)^2$$

$$0.5 = V_{GS} - V_t$$

$$V_{GS} = 1.2 V$$

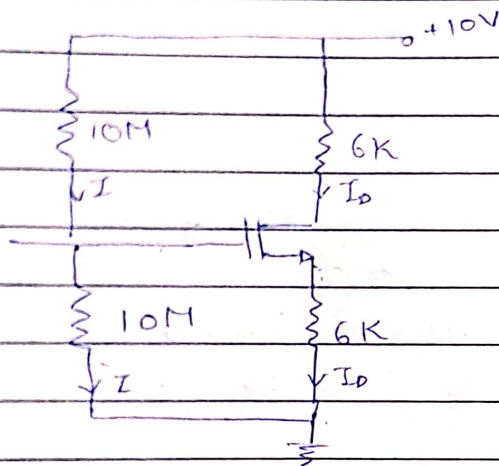
$$V_S = -1.2$$

$$V_{DS} = V_D - V_S = 0.5 - (-1.2) = 1.7$$

$$1.7 > V_{DS} - V_t \rightarrow 1.7 \geq 1.2 - 0.4, 1.7 \geq 0.5 =$$

$$R_s = \frac{V_S - (V_{DS})}{I_D} = \frac{(-0.5) - 1.2}{0.4 \times 10^{-3}} = 3.25 \text{ k}\Omega$$

HW

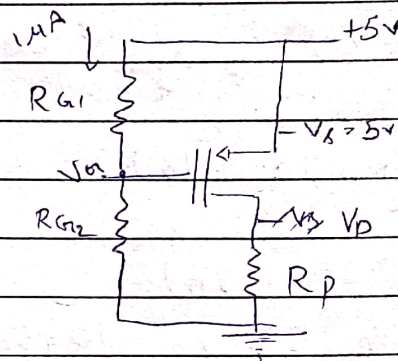


$$\mu_n C_{ox} \frac{W}{L} =$$

$$V_t = 1 \text{ V}$$

All nodal voltages & all branch currents

① Design a circuit so that transistor operates in saturation with $I_D = 0.5 \text{ mA}$, $V_D = 3 \text{ V}$, $V_t = -1 \text{ V}$, $K_p' \left(\frac{W}{L} \right) = 1 \text{ mA/V}^2$, $\lambda = 0$ ii) Write $R_{p \text{ max}}$



$$I_D = \frac{K_n'}{2} (V_{ov})^2$$

$$0.5 \text{ mA} = \frac{1 \text{ mA}}{2} (V_{ov})^2$$

$$V_{ov} = \pm \sqrt{1}$$

$$-1 \text{ V} = V_{ov}$$

$$V_{DS} = -1, \quad V_D = V_S = -1, \quad V_G = 4$$

$$V_{GS} - V_t = -1, \quad V_{GS} = -1 + V_t = -2 \text{ V}$$

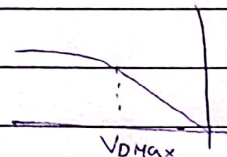
$$V_{GS} = V_G - V_S, \quad -2 = V_G - V_S, \quad V_G = 5 - 2 = 3$$

$$\frac{5 - 3}{R_{G1}} = 10^{-6}, \quad R_{G1} = 2 \text{ M}\Omega, \quad \frac{3 - 0}{R_{G2}} = 10^{-6}, \quad 3 \text{ M}\Omega = R_{G2}$$

$$R_p = \frac{V_D}{I_D} = \frac{3}{0.5 \text{ mA}} = 6 \text{ K}\Omega$$

$$\rightarrow V_{DS \text{ MAX}} = V_{GS} - V_t = -2 + 1 = -1$$

$$\rightarrow R_{p \text{ MAX}} = \frac{-1}{0.5 \text{ mA}} = 2 \text{ K}\Omega$$



$$V_{D \text{ max}} = V_G - V_t, \quad \Rightarrow V_{D \text{ max}} = 3 - (-1) = 4$$

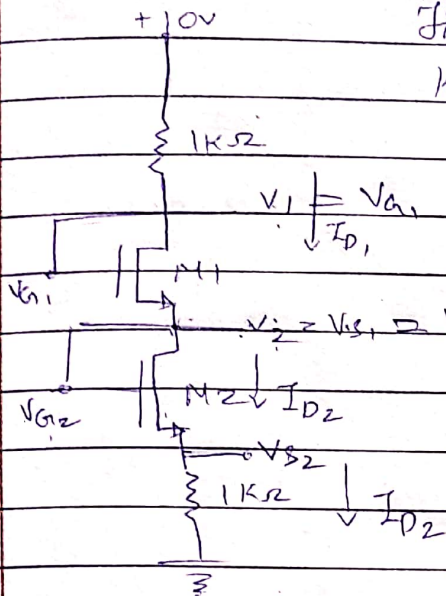
$$R_{p \text{ max}} = \frac{4}{0.5 \text{ mA}} = 8 \text{ K}\Omega$$

Gate & drain shorted $\rightarrow$ saturation

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②



Find V_1, V_2, V_3 & I_D $V_t = 1V$

$$K_n' = 2mA/V^2$$

Transistor in sat. ($V_{D1} = V_{G1}, V_{D2} = V_{G2}$)

$$I_{D1} = \frac{K_n'}{2} (V_1 - V_2 - 1)^2$$

$$V_1 = 10 - I_{D1} R_D \quad \text{--- (1)}$$

$$V_2 = R_S I_{D2} \quad \text{--- (4)}$$

$$I_{D1} = \frac{2m}{2} (V_1 - V_2 - 1)^2$$

$$I_{D1} = (V_1 - V_2 - 1) mA \quad \text{--- (2)}$$

$$I_{D2} = m (V_2 - V_3 - 1) A$$

$$V_1 = 7.56$$

$$I_{D1} = I_{D2} \Rightarrow V_1 - V_2 - 1 = V_2 - V_3 - 1$$

$$V_3 = 2.45$$

$$V_1 - 2V_2 + V_3 = 0$$

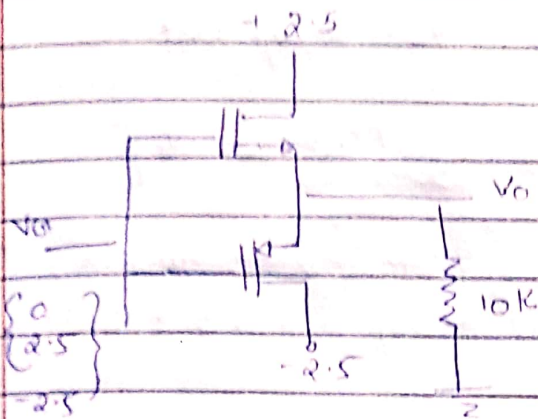
$$I_D = 2.44 mA$$

$$\frac{V_1 + V_3}{2} = V_2 \quad \text{--- (5)}$$

$$\frac{10 - I_{D1} R_D + R_S I_{D2}}{2} = V_2, \quad V_2 = \frac{10 - I_D (2K)}{2} = 5 - 1K I_D$$

Case i) $V_{G1} = 0V$

$$V_S = \{2.5 \text{ to } -2.5\}$$



NMOS

PMOS

$$V_O = V_G$$

$$\text{when } V_G = 2.5V$$

$$V_G = 2.5V$$

$$V_{GS} = -2.5 < V_t \Rightarrow \text{off} \quad V_{GS} = -2.5$$

$$V_S = -2.5$$

ON

$$V_{GS} = 2.5 > V_{th} \Rightarrow \text{ON}$$

$$V_S = -2.5$$

$$V_{GS} = 2.5 > V_{th}$$

off

$$V_{tp} = V_{th} = 1V$$

$$K_n' \frac{W}{L} = 1 \text{ mA/V}^2$$

Case ii) $V_{G1} = +2.5$

NMOS

PMOS

$$V_S = 2.5V$$

$$V_G = 2.5V$$

$$V_{GS} = 0 < V_t \Rightarrow \text{off} \quad V_{GS} = 0 > V_t \Rightarrow \text{off}$$

(or 1000)
(after value)

$$V_S = -2.5V \quad V_{GS} = 5 > V_t \Rightarrow \text{ON}$$

$$V_S = -2.5 \quad V_{GS} = 5 > V_t \Rightarrow \text{off}$$

$$V_{GS} = 0 < V_t$$

$$I_{Dn} = \frac{1}{2} \left(\mu_n C_{ox} \frac{W}{L} \right) (V_{GS} - V_t)^2, \quad I_{Dn} = \frac{1}{2} m (2.5 - V_S - 1)^2$$

$$V_S = 10 I_{Dn}, \quad I_{Dn} = \frac{1}{2} m (2.5 - 10 I_{Dn} - 1)^2 \Rightarrow 2K I_{Dn} = 1.5 - 10 I_{Dn}$$

$$(2K + 10) I_{Dn} = 1.5, \quad I_{Dn} = 0.74 \text{ mA}, \quad 2K I_{Dn} = 2.25 + 100 I_{Dn}^2 - 50 I_{Dn}$$

$$100 I_{Dn}^2 - 2050 I_{Dn} + 2.25 = 0$$

$$I_{Dn} = 0.2156 \text{ mA}, \quad I_{Dn} = 0.104 \text{ mA} \Rightarrow V_S = 10 I_{Dn}$$

$$\because V_{GS} > V_t \Rightarrow I_{Dn} = 0.104 \text{ mA}$$

Case iii) $V_{G1} = -2.5$

NMOS

PMOS

$$V_S = 2.5$$

$$V_G = 2.5$$

$$V_{GS} = -5V$$

$$V_{GS} = -5V, \quad V_{GS} < V_t \Rightarrow \text{ON}$$

$$V_{GS} < V_t \Rightarrow \text{off}$$

$$V_S = -2.5$$

$$V_S = -2.5$$

$$V_{GS} = 0$$

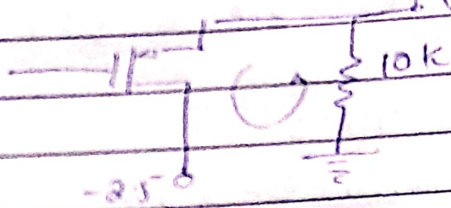
$$V_{GS} < V_t \Rightarrow \text{OFF}$$

$$V_S = 0 < V_t \Rightarrow \text{off}$$

$$V_{GS} > V_t \Rightarrow \text{Sat}$$

$$0 > V_t$$

$$V_S = -I_{DP} 10K$$



$$I_{DP} = \frac{1m}{2} (V_G - V_S - V_t)^2$$

$$I_{DP} = 0.5m (-2.5 + I_{DP} 10K + 1)^2$$

$$I_{DP} = 0.5m (10K I_{DP} - 1.5)^2 \Rightarrow I_{DP} = 2.25 + 100M I_{DP}^2 - 30000$$

$$100M I_{DP}^2 - I_{DP} - 29997.75 = 0$$

$$I_{DP} = 0.0173, -0.0173$$

$\frac{w}{L}$ - aspect ratio

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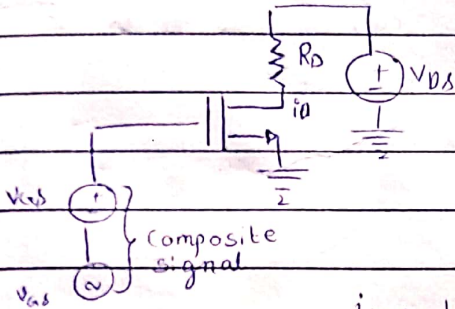
Small signal operation & Model

→ Variation of I_D wrt V_{GS}

AC composite $I = I_D + i_d$

$w \cdot k \cdot t$

$$I_D = \frac{1}{2} \mu_n C_{ox} \frac{w}{L} (V_{GS} - V_t)^2$$



$$i_D = \frac{1}{2} \mu_n C_{ox} \frac{w}{L} (V_{GS} + v_{gs} - V_t)^2$$

$$i_D = \frac{1}{2} \mu_n C_{ox} \frac{w}{L} \left[\underbrace{V_{GS} - V_{GS} V_t}_A + \underbrace{v_{gs}}_B \right]^2$$

$\xleftarrow{\quad} K_n'$

$$i_D = K_n' \left((V_{GS} - V_t)^2 + \cancel{v_{gs}^2} + 2v_{gs}(V_{GS} - V_t) \right)$$

$\ll 1$

$$i_D = K_n' (V_{GS} - V_t)^2 + 2K_n' v_{gs} (V_{GS} - V_t)$$

$$i_D = I_D + \underbrace{2K_n' v_{gs} [V_{GS} - V_t]}_{i_d}$$

$$i_d = 2 \times \frac{1}{2} \mu_n C_{ox} \frac{w}{L} [v_{gs}] [V_{GS} - V_t]$$

$$i_d = \mu_n C_{ox} \frac{w}{L} v_{gs} (V_{GS} - V_t)$$

$$\frac{i_d}{v_{gs}} = \frac{1}{r_d} = g_m = \mu_n C_{ox} \frac{w}{L} (V_{GS} - V_t) \Rightarrow \text{ac model parameter / transconductance}$$

$$g_m = 2m \times 2$$

$$\boxed{g_m = 4m}$$

$$A_D = \frac{g_m R_{ov}}{2}$$

$$g_m = 0.5 \text{ mA/V}$$

$$= 4m \times 0.03$$

$$g_m = 0.12m$$

$$A_D = 4m \times 2 = 4mV$$

$$A_D = 2.6V$$

$$A_D = \frac{1}{g_m} = 2.6k \Omega$$

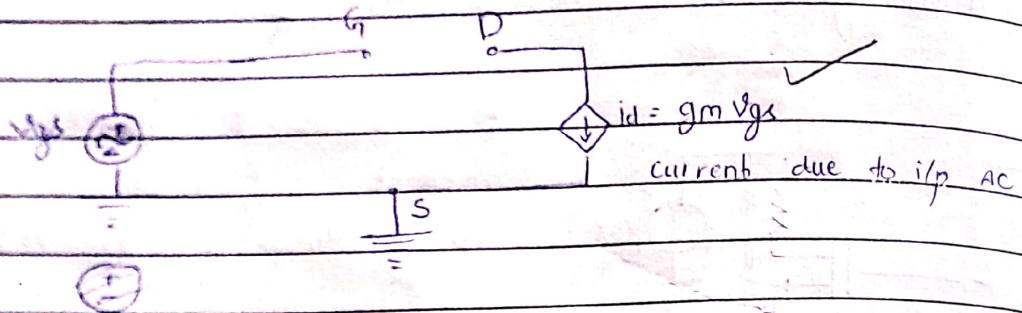
$$V_{GS} = 3$$

$$V_t = 1$$

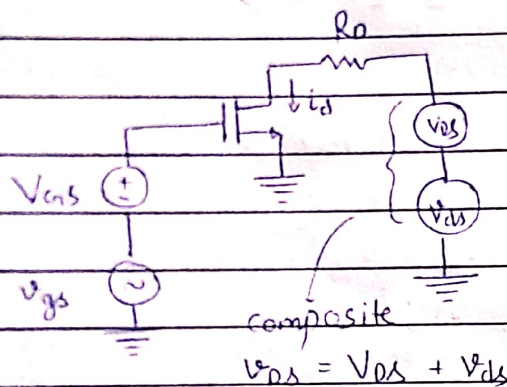
$$A_D = \frac{V_A}{R_{ov}} = \frac{30}{4m} \Rightarrow 7.5k$$

$$\lambda = \frac{1}{V_A} \approx 30$$

→ AC equivalent for MOSFET



Variation of I_D wrt V_{DS}



$$i_D = I_D + i_d$$

$$I_D = I_{Dsat} (1 + \lambda V_{DS})$$

$$i_D = I_{Dsat} (\lambda V_{DS})$$

$$i_D = I_{Dsat} (\lambda V_{DS})$$

$$i_D = I_{Dsat} (\lambda [V_{DS} + V_{DS}])$$

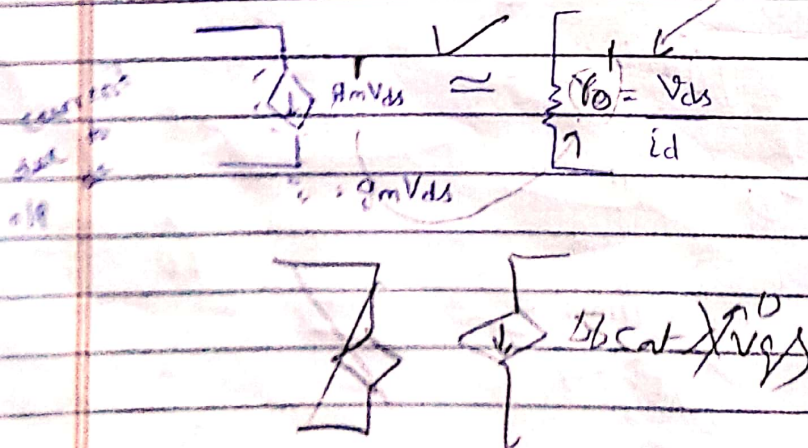
$$i_D = I_{Dsat} (1 + \lambda V_{DS} + \lambda V_{DS})$$

$$i_D = I_{Dsat} + \underbrace{I_{Dsat} \cdot \lambda \cdot V_{DS}}_{i_d}$$

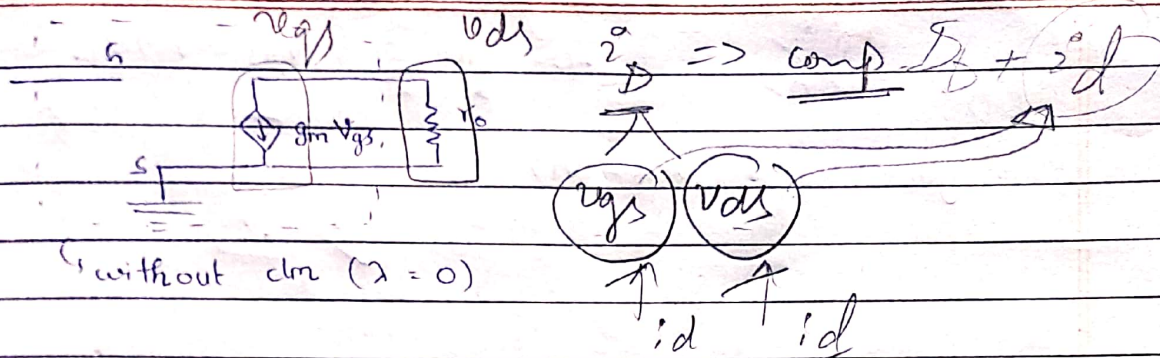
$$i_d = \frac{I_{Dsat} \lambda V_{DS}}{g_m}$$

$$i_d = g_m V_{DS}$$

$$r_o = \frac{1}{\lambda I_{Dsat}} = \frac{1}{g_m}$$



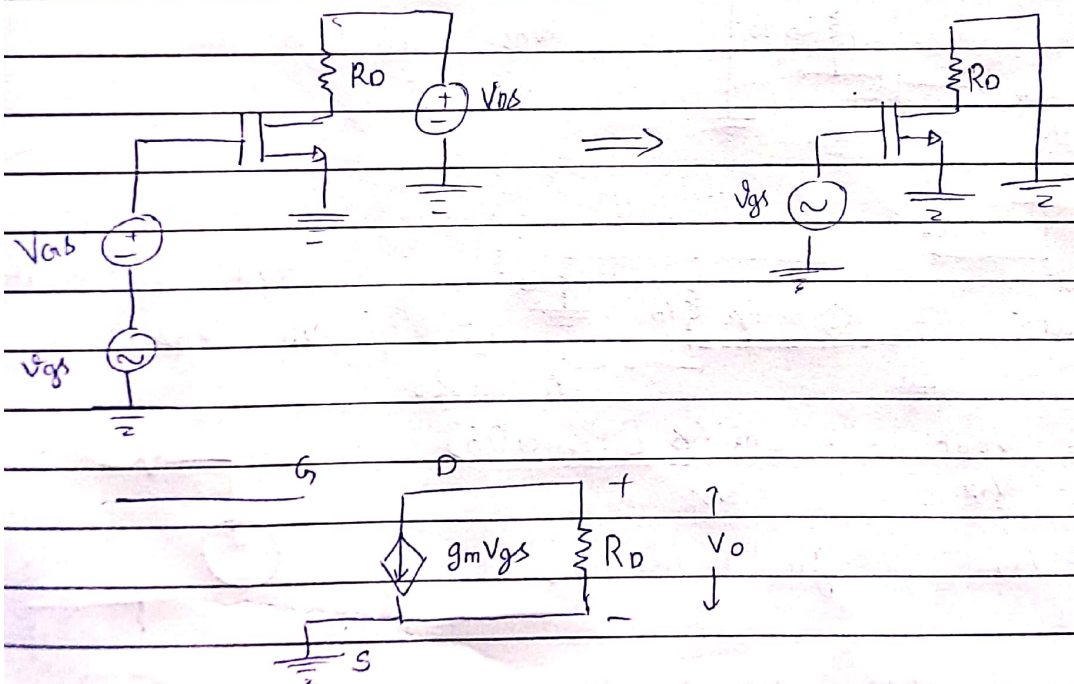
$$g_m = \frac{I_{Dsat}}{V_{DS}}$$



MOSFET amplifier configuration

	i/p	o/p
Common source amp	G	D
Gate	S	D
drain	G	S

Common Source amplifier without CM [$\lambda = 0$]



$$A_v = \frac{v_{out}}{v_{in}} = - \frac{i_d R_D}{v_{gs}} = - \frac{g_m v_{gs} R_D}{v_{gs}} = -g_m R_D$$

$$R_{in} = \frac{v_{in}}{i_{in}} = \frac{v_{in}}{0} = \infty \quad R_{out} = \frac{v_{out}}{i_{out}} = \frac{v_{out}}{0} = \infty \quad (\text{Keep } v_{in} = 0 \text{ supply } v \text{ at o/p})$$

Input AC current is very small $\therefore$ we neglect that & $i_{in} = 0$

$$g_m = \frac{2 I_D}{V_{ov}} \quad , \quad g_m = \sqrt{2 I_D \mu_n C_{ox} \frac{\omega}{L}}$$

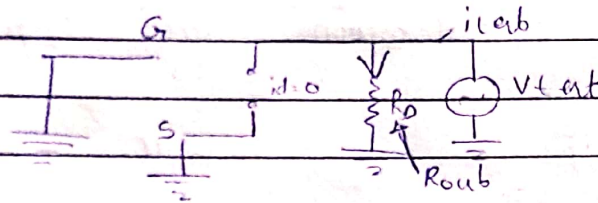
$$g_m = \mu_n C_{ox} \frac{\omega}{L} (V_{ovs} - V_t) \quad (3)$$

$$I_D = \frac{g_m V_{ov}}{2}$$

$$I_D = \frac{K_n' \omega}{2} (V_{ovs} - V_t) V_{ovs}$$

$$g_m = K_n' \omega (V_{ovs} - V_t)$$

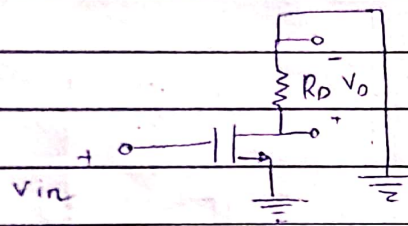
for R_{out} set $v_{in} = 0$, $V_{gs} = 0$, $i_d = 0$



$$R_{out} = R_D$$

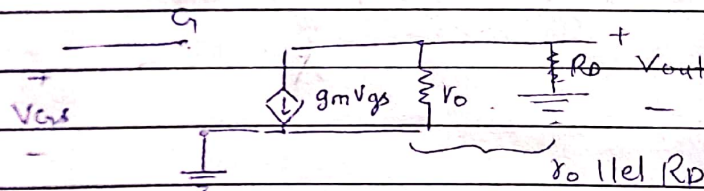
$$R_{out} = \infty \parallel R_D = R_D$$

→ Common Source amplifier with CLM [$\lambda \neq 0$]



V_{gs} set to '0'

V_{gs} set to '0'



$$V_{out} = i_d (R_D \parallel R_L)$$

$$V_{in} = V_{gs}$$

$$A_v = \frac{V_{out}}{V_{in}} = - \frac{g_m V_{gs}}{V_{gs}} [R_D \parallel R_L]$$

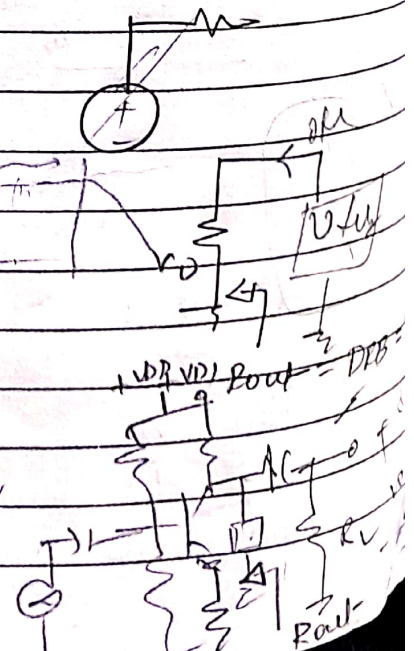
$$A_v = -g_m [R_D \parallel R_L]$$

$$R_{in} = \frac{V_{in}}{i_{in}} = \frac{V_{in}}{0} = \infty$$

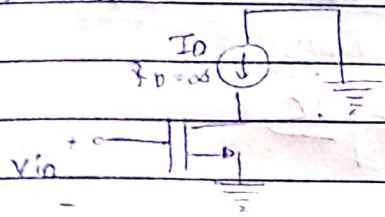
$$R_{out} = \frac{V_{out}}{i_{out}} = (R_D \parallel R_L)$$

DB

DB 50'



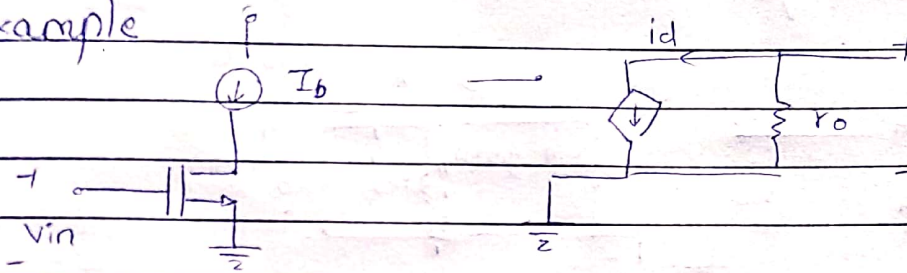
If we don't bias with R_D , we can put current source



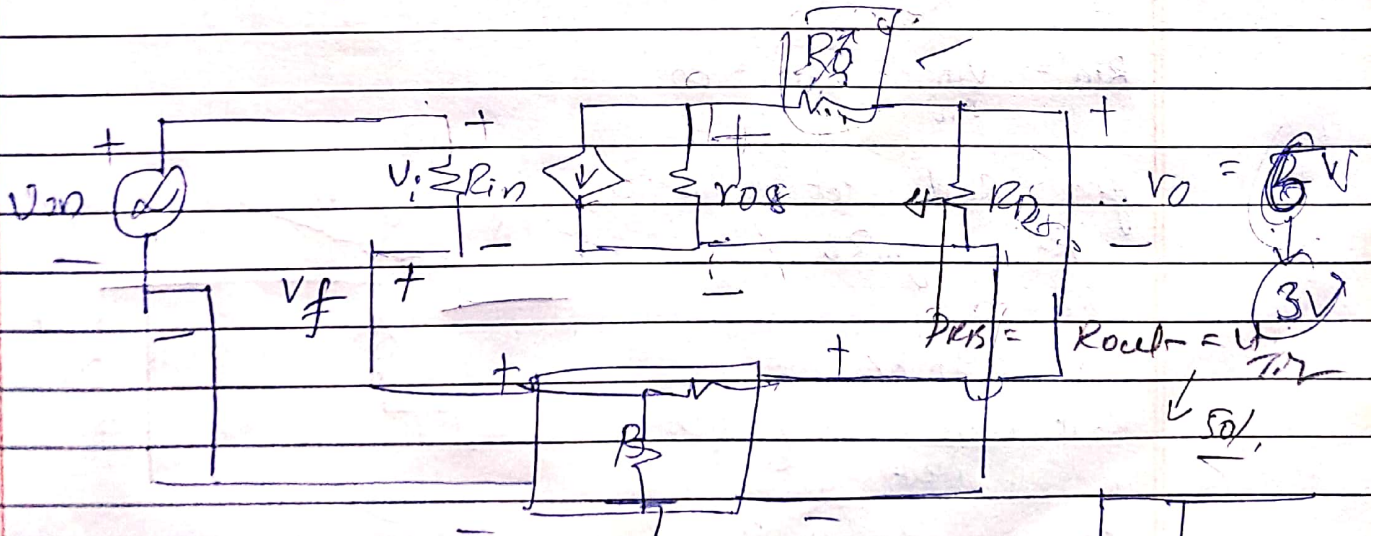
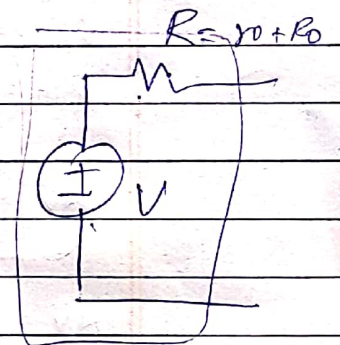
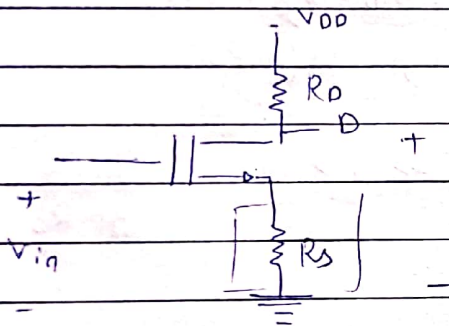
$[\lambda \neq 0]$

Practically we have current mirrors for current source

Example

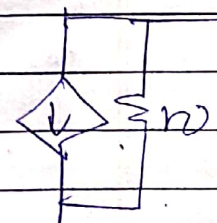


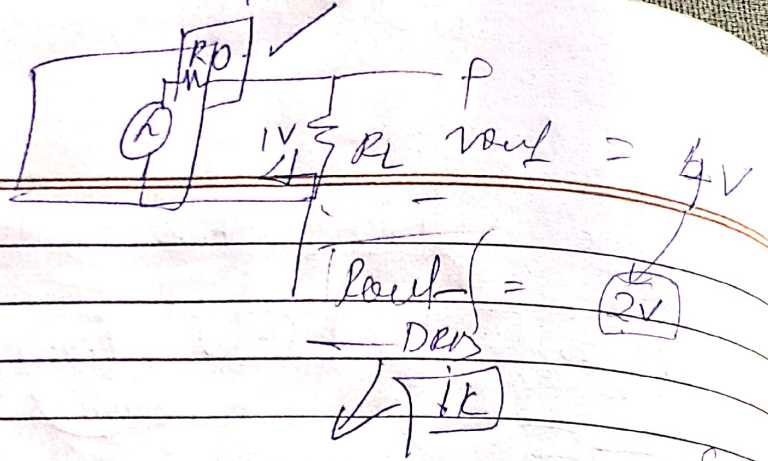
Common source with R_S [source resistance]



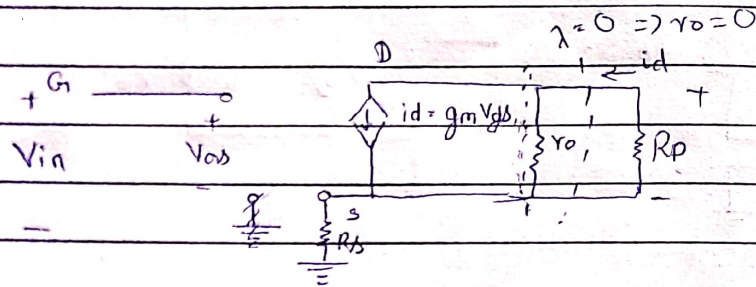
$$v_{in} - v_i = v_f = 0$$

$$v_{on} = v_f = v_i$$





→ AC equivalent for CS with R_S ($\lambda = 0$)



$$A_v = \frac{V_{out}}{V_{in}} \quad V_{out} = -i_d R_D = -g_m V_{gs} R_D$$

$$V_{in} = V_{gs} + V_s \text{ (across source)} \Rightarrow V_{in} = V_{gs} + i_d R_S$$

$$V_{in} = V_{gs} + g_m V_{gs} R_S, \quad V_{in} = V_{gs} (1 + g_m R_S)$$

$$A_v = \frac{-g_m R_D}{(1 + g_m R_S)} \quad [as \quad g_m R_S \gg 1] \Rightarrow A_v = -\frac{g_m R_D}{g_m R_S}$$

$$A_v = -\frac{R_D}{R_S}$$

$$R_{in} = \frac{V_{in}}{I_{in}} = \frac{V_{in}}{0} = \infty$$

for R_{out} let $V_{in} = 0$, $V_{in} = V_g + V_s \Rightarrow V_{gs} = -V_s$

$$R_{in} = -g_m R_D$$

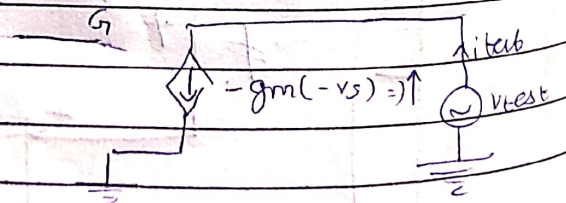
$$R_{in} = 0$$

$$A_v = -g_m R_S$$

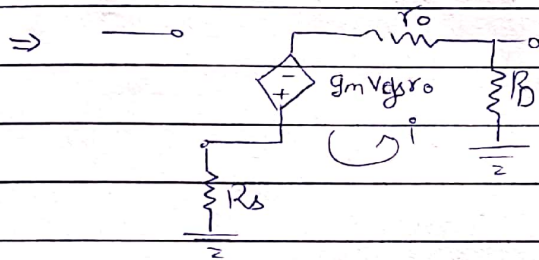
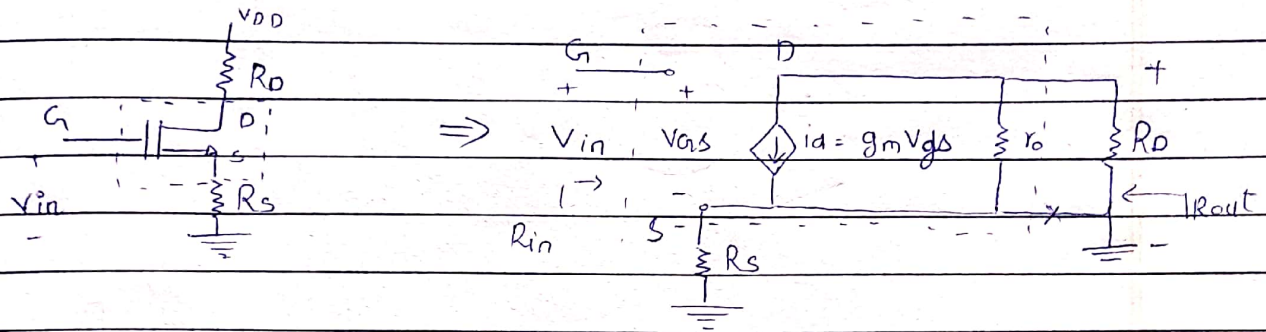
$$R_{out} = \frac{V_{test}}{i_{test}} = \frac{V}{0} = \infty$$

$$R_{out} = \frac{R_{out} \parallel R_D}{1}$$

$$= R_D$$



→ CS with R_S [$\lambda = 0$]



$$g_m V_{gs} r_o = (R_S + r_o + R_D) i$$

$$i = \frac{g_m V_{gs} r_o}{(R_S + r_o + R_D)}$$

$$V_{out} = -i R_D \Rightarrow V_{out} = -\frac{g_m V_{gs} r_o R_D}{(R_S + r_o + R_D)}$$

$$V_{in} = V_{gs} + V_S \quad V_{in} = V_{gs} + \frac{g_m V_{gs} r_o R_S}{R_S + r_o + R_D}$$

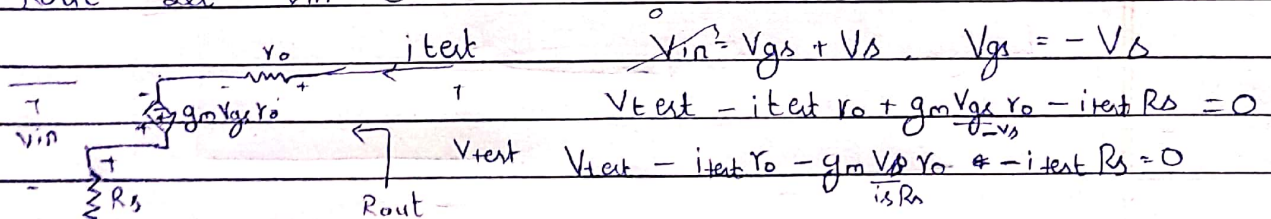
$$V_{in} = V_{gs} (R_S + r_o + R_D + \frac{g_m r_o R_S}{R_S + r_o + R_D})$$

$$A_v = \frac{V_{out}}{V_{in}} = -\frac{g_m V_{gs} r_o R_D}{R_S + r_o + R_D} \cdot \frac{1}{V_{gs} (R_S + r_o + R_D + \frac{g_m r_o R_S}{R_S + r_o + R_D})}$$

$$A_v = \frac{-g_m R_D}{(1 + \frac{R_S}{r_o} + \frac{R_D}{r_o} + g_m V_{gs} R_S)}$$

when $r_o = \infty$ $A_v = -\frac{g_m R_D}{(1 + g_m R_S)}$ when $R_S = 0$, $A_v = -g_m R_D$

R_{out} : set $V_{in} = 0$



$$V_{test} = i_{test} r_o + g_m r_o i_{test} R_S + i_{test} R_D$$

$$V_{test} = i_{test} (r_o + g_m r_o R_S + R_D)$$

$$R_{out}' = R_{out} \parallel R_D$$

$$R_{out}' = (g_m r_o R_S + R_S + r_o) \parallel R_D$$